John P. Acosta

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY *Newark, NJ*

Bachelor of Science in Computer Science, Ying Wu College of Computing **Expected Dec 2027**

- GPA: 3.7 **- Awards**: NSF S-STEM Scholar, Dean's List (All Semesters), Ron Andrews PGIM Scholar, Kupfrian Memorial Scholar

- Relevant Coursework: Operating Systems, Computer Systems, Computer Networks, Programming Language Concepts (C/C++), Data Structures & Algorithms, Object-Oriented Programming, Database Design & Management, Artificial Intelligence

SKILLS

Languages: Python, C/C++, Java, JavaScript, Bash
Systems & Infrastructure: Linux, Docker, Terraform, SLURM/HPC, AWS, Azure
Build, CI/CD & Development: GitHub Actions, Git/GitHub, pytest
AI/ML & Frameworks: PyTorch, Transformers, PEFT, LangChain, FastAPI, React/Next.js, PostgreSQL, SQLite

PROFESSIONAL EXPERIENCE

Software Engineer Intern **Jun 2026 - Aug 2026**

PGIM Technology DevOps

- Developed 7 reusable Terraform modules for AWS Lambda, S3, and KMS, reducing average application build time by ~66% and infrastructure code by ~80% while migrating from CloudFormation under enterprise IaC standards spanning 7 business units
- Automated cloud cost reporting for $1M+ in monthly spend through a React, FastAPI, and AWS S3 dashboard, replacing a 3-hour manual monthly reporting process
- Led a 5-engineer team to 2nd place in PGIM's internal hackathon, prototyping an AI-powered Confluence documentation pipeline for stale-page detection, update suggestions, and reviewer notifications to improve knowledge quality for Copilot and Claude MCP

Undergraduate Research Assistant **Jan 2026 - Jun 2026**

Lijing Wang Lab | NJIT

- Tracked down intermittent LLM fine-tuning variance to RNG inconsistencies in data ordering, weight initialization, and dropout; implemented per-factor seed controls, preventing invalidation of 400+ planned runs
- Designed experiments across Llama 3.2-3B, RoBERTa, and BERT, scheduling GPU workloads through SLURM on NJIT's Wulver A100 HPC cluster

Applied AI Research Engineer Intern **Jan 2026 - Jun 2026**

NJIT AIM Lab

- Ported a neurosurgical PyQt application to Linux HPC and developed an automated QA harness covering 100 natural-language prompt suites, capturing UI screenshots and execution telemetry to isolate failures and guide fixes that raised test pass rates from 20% to 100%
- Prototyped LangChain workflows translating surgeon-issued natural-language instructions into tumor segmentation tool calls; analyzed BraTS imaging datasets to inform model selection and presented findings to biomedical researchers

Deep Learning Forecasting Research Intern **Aug 2025 - Jan 2026**

Grace Hopper AI Research @ NJIT

- Parallelized independent forecasting workloads across CPU cores using joblib multiprocessing, reducing experiment runtime by ~80% and enabling faster iteration without HPC resources
- Built fault-tolerant checkpointing and logging utilities to recover interrupted experiments without redundant computation, automating metric collection and result aggregation
- Engineered modular Python preprocessing routines (NumPy, pandas) for cleaning, trimming, and interpolating 10+ multi-location epidemic datasets (~900 weekly observations each), creating consistent train/test splits and enabling reliable rolling-origin forecasting across 5 disease types

TECHNICAL PROJECTS

SHPE NJIT APP | Developer | GitHub Actions, Supabase

- Own CI/CD for the SHPE NJIT chapter mobile app, architecting automated test and deploy pipelines via GitHub Actions to support 200+ active members

AI-TUTOR | Lead Developer | Python, GraphRAG, Docker, Flask, BeautifulSoup

- Built a knowledge-graph-based tutoring system using a custom BFS web crawler and an extended nano-graphrag implementation with domain-specific entity extraction to ground LLM responses in retrieved context
- Implemented rate-limited LLM requests with exponential backoff, integrating a Flask backend and Streamlit interface within a Docker development environment supporting selective service loading

EXTRACURRICULAR ACTIVITIES

Society of Hispanic Professional Engineers | NJIT | ATTENDING SHPE NATIONAL CONVENTION 2026 **Jan 2024 - Present**
HISPA (Hispanics Inspiring Students' Performance and Achievements) | Member & Role Model **Sep 2025 - Present**
Break Through Tech AI Fellow | Cornell Tech **Jun 2025 - Present**
Google Software Engineering Fellow (Selected 2x) | G-SWEP **Jul 2025 - Sep 2025**